

Qualification Pack



Green Hydrogen Plant Supervisor

QP Code: HYC/Q4003

Version: 1.0

NSQF Level: 5

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HYC/Q4003: Green Hydrogen Plant Supervisor

Brief Job Description

A Green Hydrogen Plant Supervisor is responsible for leading the team in operations, maintenance and safety of a green hydrogen production facility. The person at this job has deep understanding of the mechanical and electrical principles and processes involved in air separation and hydrogen generation. His role is to manage the seamless operation of critical equipment like electrolyzers, dryers, deoxy reactor, electrical heater, pump, compressors, and storage systems, ensuring smooth production processes. Adherence to process Safety and safety protocol during the production process is also one of the primary responsibilities of the Plant Supervisor

Personal Attributes

The individual should have ability to give attention to detail, technical aptitude, adaptability, leadership teamwork and management and a strong commitment to safety and sustainability. The person should have the ability to work individually / team round the clock and must always promote adherence to safety protocol.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HYC/N4001: Hydrogen Safety, Security and Health Management Procedures](#)
2. [HYC/N4006: Hydrogen Technology Overview and Operations Management](#)
3. [HYC/N4007: Managing Hydrogen production system & Technologies](#)
4. [HYC/N9301: Working effectively in a team](#)
5. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Hydrogen Production, Storage & Transportation
Country	India
NSQF Level	5

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Credits	22
Aligned to NCO/ISCO/ISIC Code	NCO 2015/1321.0600
Minimum Educational Qualification & Experience	Completed 3 year UG degree with 1 Year of experience OR 12th grade pass with 2 year NTC/ CITS/NAC with 2 Years of experience OR Completed 2nd year diploma after 12th with 2 Years of experience OR 12th grade Pass with 4 Years of experience OR Previous relevant Qualification of NSQF Level (4.5) with 1-2 Years of experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	29/11/2026
NSQC Approval Date	30/11/2023
Version	1.0
Reference code on NQR	QG-05-HY-01364-2023-V1-HSSCI
NQR Version	1.0

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HYC/N4001: Hydrogen Safety, Security and Health Management Procedures

Description

This unit covers maintaining health, safety and security procedures at the workplace where Hydrogen is produced and handled.

Scope

The scope covers the following :

- Understanding Properties of Hydrogen from safety point of view
- Safe handling of Hydrogen
- Follow fire safety procedures
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Understanding Properties of Hydrogen from safety point of view & Safe handling of Hydrogen

To be competent, the user/individual on the job must be able to:

- PC1.** Ensure that personnel under their supervision have a deep understanding of hydrogen's properties, including its flammability, explosive limits and potential hazards.
- PC2.** Ensure personal and team adherence to the use of appropriate protective clothing/equipment for specific tasks and work conditions
- PC3.** Identify and communicate the relevant documents, responsible individuals, and designated locations pertaining to health and safety in the workplace, facilitating effective safety management.
- PC4.** Oversee and participate in risk assessments specific to hydrogen storage, handling and production processes.
- PC5.** Proficiently identify hydrogen process hazards and assess risks including the analysis of risk or accident in the workplace due to hydrogen leak/fire
- PC6.** capability to accurately read and interpret data from sensing equipment used in hydrogen production processes
- PC7.** apply key safety parameters for handling hydrogen & oxygen
- PC8.** Evaluate hydrogen release pressures and understand hydrogen flame detection methods, emphasizing the colorless nature of hydrogen flames
- PC9.** Apply critical safety requirements for electrical equipment used in green hydrogen production, minimizing electrical hazards
- PC10.** competent in the operation and functioning of hydrogen exhaust systems, facilitating the safe removal of hydrogen from the workplace
- PC11.** Execute inertization processes using nitrogen to displace hazardous gases enhancing safety in hydrogen-related operations
- PC12.** Proficiently identify and recognize common safety signs displayed in various areas, promoting a safe work environment.

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- PC13.** safely handle and manage lye solution (KOH solution), ensuring the prevention of accidents and injuries
- PC14.** carry out safe working practices while dealing with hazards to ensure the safety of self and others
- PC15.** Enforce the use of safety guidelines and procedures that consider hydrogen's unique properties.
- PC16.** Competently manage hydrogen ignition sources to prevent potential incidents and enhance workplace safety.
- PC17.** Adhere to standard practices for the safe venting of hydrogen and oxygen, referencing applicable industry standards and performance requirements for equipment and hoses, ensuring safe gas handling procedures.

Ensure the following fire safety procedures

To be competent, the user/individual on the job must be able to:

- PC18.** Ensure the personal and team Familiarization with Hydrogen Leak and flame detection system, Identification of Hydrogen Fire; use the various appropriate fire extinguishers on different types of fires correctly
- PC19.** Develop and implement measures to mitigate hydrogen fire, isolation of Hydrogen supply line and lead the team in following rescue techniques applied during fire hazard
- PC20.** Promote and enforce following good housekeeping practice in order to prevent fire hazards
- PC21.** Regularly inspect work areas for compliance with housekeeping standards
- PC22.** identify and document issues concerning the hydrogen safety in work place
- PC23.** Collaborate with relevant departments to address and rectify safety concerns promptly
- PC24.** Ensure that work areas and escape routes are well maintained and free from any hindrance
- PC25.** Establish and communicate a system for personnel to report any near-miss incidents to the fire safety department and provide the necessary guidance and resources for reporting and investigation
- PC26.** Lead by example in consistently following and enforcing the applicable laws, regulations and codes as per safety standard
- PC27.** Ensure that personnel are trained in preparing comprehensive accident/incident reports
- PC28.** Review and approve written accident/incident reports before sharing them with the concerned officer or department to facilitate proper documentation and follow-up actions

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC29.** provide appropriate first aid to victims during emergency situations, ensuring prompt and effective response to injuries or medical issues
- PC30.** Be proficient in basic techniques of bandaging
- PC31.** Lead by example in responding promptly and appropriately to an accident
- PC32.** perform rescue activity during an accident in real or simulated environments
- PC33.** Familiarize personnel with correct escape routes during an emergency
- PC34.** demonstrate correct method to rescue injured people and others during an emergency
- PC35.** Ensure that personnel are familiar with emergency protocols, including evacuation and shutdown procedures and verify that the team understands and can implement these protocols effectively in case of emergencies.

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company policies on personnel management and duty reporting procedure
- KU2.** reporting structure within organization
- KU3.** how to escalate problem
- KU4.** concept of hazards and risks
- KU5.** health and safety hazards commonly affecting the work environment and related precautions
- KU6.** preventative and remedial actions to be taken in the case of exposure to toxic materials
- KU7.** importance of using protective clothing/equipment while working
- KU8.** various causes of fire
- KU9.** techniques of using different types of fire extinguishers
- KU10.** different materials used for extinguishing fire
- KU11.** various types of safety signs and their significance
- KU12.** the concept of Mission Karmayogi
- KU13.** the concept of Swachhata Hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate by writing
- GS2.** read instructions, guidelines/procedures and reports
- GS3.** communicate effectively and share the information efficiently
- GS4.** identify and report potential sources of danger
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of on time services
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding Properties of Hydrogen from safety point of view & Safe handling of Hydrogen</i>	17	27	-	-
PC1. Ensure that personnel under their supervision have a deep understanding of hydrogen's properties, including its flammability, explosive limits and potential hazards.	1	2	-	-
PC2. Ensure personal and team adherence to the use of appropriate protective clothing/equipment for specific tasks and work conditions	1	1	-	-
PC3. Identify and communicate the relevant documents, responsible individuals, and designated locations pertaining to health and safety in the workplace, facilitating effective safety management.	1	2	-	-
PC4. Oversee and participate in risk assessments specific to hydrogen storage, handling and production processes.	1	2	-	-
PC5. Proficiently identify hydrogen process hazards and assess risks including the analysis of risk or accident in the workplace due to hydrogen leak/fire	1	2	-	-
PC6. capability to accurately read and interpret data from sensing equipment used in hydrogen production processes	1	2	-	-
PC7. apply key safety parameters for handling hydrogen & oxygen	1	2	-	-
PC8. Evaluate hydrogen release pressures and understand hydrogen flame detection methods, emphasizing the colorless nature of hydrogen flames	1	1	-	-
PC9. Apply critical safety requirements for electrical equipment used in green hydrogen production, minimizing electrical hazards	1	1	-	-
PC10. competent in the operation and functioning of hydrogen exhaust systems, facilitating the safe removal of hydrogen from the workplace	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Execute inertization processes using nitrogen to displace hazardous gases enhancing safety in hydrogen-related operations	1	1	-	-
PC12. Proficiently identify and recognize common safety signs displayed in various areas, promoting a safe work environment.	1	1	-	-
PC13. safely handle and manage lye solution (KOH solution), ensuring the prevention of accidents and injuries	1	2	-	-
PC14. carry out safe working practices while dealing with hazards to ensure the safety of self and others	1	2	-	-
PC15. Enforce the use of safety guidelines and procedures that consider hydrogen's unique properties.	1	2	-	-
PC16. Competently manage hydrogen ignition sources to prevent potential incidents and enhance workplace safety.	1	1	-	-
PC17. Adhere to standard practices for the safe venting of hydrogen and oxygen, referencing applicable industry standards and performance requirements for equipment and hoses, ensuring safe gas handling procedures.	1	1	-	-
<i>Ensure the following fire safety procedures</i>	11	19	-	-
PC18. Ensure the personal and team Familiarization with Hydrogen Leak and flame detection system, Identification of Hydrogen Fire; use the various appropriate fire extinguishers on different types of fires correctly	1	2	-	-
PC19. Develop and implement measures to mitigate hydrogen fire, isolation of Hydrogen supply line and lead the team in following rescue techniques applied during fire hazard	1	2	-	-
PC20. Promote and enforce following good housekeeping practice in order to prevent fire hazards	1	2	-	-
PC21. Regularly inspect work areas for compliance with housekeeping standards	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify and document issues concerning the hydrogen safety in work place	1	2	-	-
PC23. Collaborate with relevant departments to address and rectify safety concerns promptly	1	2	-	-
PC24. Ensure that work areas and escape routes are well maintained and free from any hindrance	1	2	-	-
PC25. Establish and communicate a system for personnel to report any near-miss incidents to the fire safety department and provide the necessary guidance and resources for reporting and investigation	1	2	-	-
PC26. Lead by example in consistently following and enforcing the applicable laws, regulations and codes as per safety standard	1	1	-	-
PC27. Ensure that personnel are trained in preparing comprehensive accident/incident reports	1	1	-	-
PC28. Review and approve written accident/incident reports before sharing them with the concerned officer or department to facilitate proper documentation and follow-up actions	1	1	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	7	14	-	-
PC29. provide appropriate first aid to victims during emergency situations, ensuring prompt and effective response to injuries or medical issues	1	2	-	-
PC30. Be proficient in basic techniques of bandaging	1	2	-	-
PC31. Lead by example in responding promptly and appropriately to an accident	1	2	-	-
PC32. perform rescue activity during an accident in real or simulated environments	1	2	-	-
PC33. Familiarize personnel with correct escape routes during an emergency	1	2	-	-
PC34. demonstrate correct method to rescue injured people and others during an emergency	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC35. Ensure that personnel are familiar with emergency protocols, including evacuation and shutdown procedures and verify that the team understands and can implement these protocols effectively in case of emergencies.	1	2	-	-
NOS Total	35	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4001
NOS Name	Hydrogen Safety, Security and Health Management Procedures
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	4.5
Credits	4
Version	2.0
Last Reviewed Date	30/04/2024
Next Review Date	29/04/2027
NSQC Clearance Date	30/04/2024

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HYC/N4006: Hydrogen Technology Overview and Operations Management

Description

This unit is about gaining a thorough understanding of various aspects related to hydrogen technology e.g., production, storage, transportation and utilization. This knowledge is essential for effective decision-making, overseeing operations and ensuring the safety and efficiency of the hydrogen production and storage facility.

Scope

The scope covers the following :

- Manufacturing (Electrolysis Plants)
- Transportation (Hazardous Materials Transportation e.g., road transportation and Pipelines transportation)
- Storage Technologies e.g. storage in gaseous and liquid form
- Utilization in domestic and industrial set up including Fuel Cell Systems, Gas Engine, and Gas Turbine Systems
- Hydrogen use in vehicles (direct injection and fuel cell)
- Hydrogen refilling station design and layout, Hydrogen Re-fuelling Systems parameters like pressure, temperature at the fuelling station, refilling parameters and equipment. Safety at the Filling station
- Handling leaks in the refilling station and handling general emergency in the refilling stations

Elements and Performance Criteria

Manufacturing (Electrolysis Plants)

To be competent, the user/individual on the job must be able to:

- PC1.** Familiarization with Engineering standards and guidelines related to production, storage, transportation and utilization of Hydrogen
- PC2.** Familiarization with types of electrolyzers and their advantages and disadvantages
- PC3.** Apply knowledge of production processes in electrolysis plants to optimize production, taking into consideration quality and safety standards.
- PC4.** Analyze and assess the production processes in electrolysis plants to identify areas for process improvement and operational efficiency.
- PC5.** Develop strategies to manage and ensure the quality of equipment used in electrolysis plants, adhering to safety regulations and compliance
- PC6.** check the Importance of Feedwater quality and maintain its quality control
- PC7.** Handle Feed water (to electrolyser) management

Transportation (Hazardous Materials Transportation and Pipelines)

To be competent, the user/individual on the job must be able to:

- PC8.** Familiarization with mode of transportation of Hydrogen e.g. gaseous Hydrogen, Liquid Hydrogen,
- PC9.** Familiarization of process of liquification of Hydrogen, comparative advantage and disadvantages of liquid hydrogen and gaseous hydrogen in transportation

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- PC10.** Analyze transportation practices related to hydrogen, evaluating safety regulations and recommending improvements for compliance
- PC11.** Develop and implement strategies for safely transporting hydrogen to and from the facility, considering regulatory requirements and efficient logistics
- PC12.** Road Transportation of Hydrogen through tube trailers / cylinders in gaseous form and road transportation of Hydrogen in liquid form
- PC13.** Pipeline transportation of hydrogen and the safety issues including suitability of utilizing existing natural gas pipelines for hydrogen transportation.
- PC14.** Apply understanding of the complexities of transporting hydrogen as a hazardous material and through road and pipelines to develop efficient logistics and safety protocols

Storage Technologies

To be competent, the user/individual on the job must be able to:

- PC15.** Familiarization with available Hydrogen storage technologies and the emerging storage technologies
- PC16.** Storing Hydrogen in Cylinders, bullets and salt caverns advantages and disadvantages
- PC17.** Selection of most appropriate storage solutions, considering safety, maintenance and utilization.
- PC18.** Analyze the performance of various hydrogen storage technologies and propose enhancements to optimize plant operations and safety.
- PC19.** Develop strategies for the safe handling of hydrogen storage, facilities ensuring efficient safety management and compliance with safety/ regulatory standards/ guidelines

Consumption (Fuel Cell Systems, Gas Engine, and Gas Turbine Systems)

To be competent, the user/individual on the job must be able to:

- PC20.** Ensuring quality of hydrogen required for various applications.
- PC21.** Apply understanding of hydrogen consumption in various applications, optimizing hydrogen utilization, monitoring consumption pattern and rate
- PC22.** Blending Hydrogen in natural gas distribution pipelines, monitor quality of blend and its safe use at homes.
- PC23.** Suitability of home burners/ industrial burners for use of hydrogen blends and safety during use of hydrogen natural gas blends at homes/ industries
- PC24.** Educating hydrogen / hydrogen blend users on correct use of burners and related safety aspect
- PC25.** Develop strategies for the effective resource allocation in hydrogen consumption applications, considering performance and operational efficiency

Hydrogen Re-fueling Systems

To be competent, the user/individual on the job must be able to:

- PC26.** Apply knowledge of hydrogen re-fueling systems to manage and maintain these systems safely and effectively, ensuring compliance with safety protocols and overseeing refueling operations
- PC27.** Analyze and assess the safety and efficiency of hydrogen re-fueling systems including storage at the refueling stations, making improvements as needed and ensuring compliance with safety standards.
- PC28.** Develop strategies for overseeing refueling processes, ensuring safety and efficiency in hydrogen re-fueling operations and compliance with relevant protocols.

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- PC29.** Monitoring parameters like pressure, temperature during refueling operation
- PC30.** Monitoring Hydrogen detector / flame detectors and activate safety protocol in case of leakage/ emergency.
- PC31.** Ensure adherence to appropriate dress code by the operating personnel at the refueling station

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of calibration of tools and instruments used in Green Hydrogen production
- KU2.** organizational procedures for safe handling of equipment and machine operations
- KU3.** reporting protocol and documentation required
- KU4.** relevant health and safety requirements applicable to the workplace
- KU5.** the importance of personal protective equipment used during the procedure performing
- KU6.** ways of efficiently managing renewable sources such as wind, water in the process
- KU7.** latest methods of energy and material conservation
- KU8.** categorization of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU9.** the difference between recyclable/reusable and waste material
- KU10.** the concept of Mission Karmayogi
- KU11.** the concept of Swachhata Hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret the test data
- GS2.** analyze the parameters and output
- GS3.** record data on waste disposal at workplace
- GS4.** assess complexity and criticality of task
- GS5.** Handle effective communicate with team members
- GS6.** perform basic mathematical calculation
- GS7.** complete tasks efficiently and accurately within stipulated time
- GS8.** work constructively and collaboratively with others.
- GS9.** delegate tasks to improve efficiencies
- GS10.** plan and organize the marketing activity

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Manufacturing (Electrolysis Plants)</i>	7	8	-	-
PC1. Familiarization with Engineering standards and guidelines related to production, storage, transportation and utilization of Hydrogen	1	-	-	-
PC2. Familiarization with types of electrolyzers and their advantages and disadvantages	1	-	-	-
PC3. Apply knowledge of production processes in electrolysis plants to optimize production, taking into consideration quality and safety standards.	1	2	-	-
PC4. Analyze and assess the production processes in electrolysis plants to identify areas for process improvement and operational efficiency.	1	2	-	-
PC5. Develop strategies to manage and ensure the quality of equipment used in electrolysis plants, adhering to safety regulations and compliance	1	2	-	-
PC6. check the Importance of Feedwater quality and maintain its quality control	1	-	-	-
PC7. Handle Feed water (to electrolyser) management	1	2	-	-
<i>Transportation (Hazardous Materials Transportation and Pipelines)</i>	10	7	-	-
PC8. Familiarization with mode of transportation of Hydrogen e.g. gaseous Hydrogen, Liquid Hydrogen,	2	-	-	-
PC9. Familiarization of process of liquification of Hydrogen, comparative advantage and disadvantages of liquid hydrogen and gaseous hydrogen in transportation	1	1	-	-
PC10. Analyze transportation practices related to hydrogen, evaluating safety regulations and recommending improvements for compliance	2	-	-	-
PC11. Develop and implement strategies for safely transporting hydrogen to and from the facility, considering regulatory requirements and efficient logistics	2	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Road Transportation of Hydrogen through tube trailers / cylinders in gaseous form and road transportation of Hydrogen in liquid form	1	2	-	-
PC13. Pipeline transportation of hydrogen and the safety issues including suitability of utilizing existing natural gas pipelines for hydrogen transportation.	1	2	-	-
PC14. Apply understanding of the complexities of transporting hydrogen as a hazardous material and through road and pipelines to develop efficient logistics and safety protocols	1	2	-	-
<i>Storage Technologies</i>	6	5	-	-
PC15. Familiarization with available Hydrogen storage technologies and the emerging storage technologies	1	1	-	-
PC16. Storing Hydrogen in Cylinders, bullets and salt caverns advantages and disadvantages	1	1	-	-
PC17. Selection of most appropriate storage solutions, considering safety, maintenance and utilization.	1	1	-	-
PC18. Analyze the performance of various hydrogen storage technologies and propose enhancements to optimize plant operations and safety.	1	2	-	-
PC19. Develop strategies for the safe handling of hydrogen storage, facilities ensuring efficient safety management and compliance with safety/ regulatory standards/ guidelines	2	-	-	-
<i>Consumption (Fuel Cell Systems, Gas Engine, and Gas Turbine Systems)</i>	6	9	-	-
PC20. Ensuring quality of hydrogen required for various applications.	-	1	-	-
PC21. Apply understanding of hydrogen consumption in various applications, optimizing hydrogen utilization, monitoring consumption pattern and rate	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. Blending Hydrogen in natural gas distribution pipelines, monitor quality of blend and its safe use at homes.	1	2	-	-
PC23. Suitability of home burners/ industrial burners for use of hydrogen blends and safety during use of hydrogen natural gas blends at homes/ industries	1	2	-	-
PC24. Educating hydrogen / hydrogen blend users on correct use of burners and related safety aspect	1	1	-	-
PC25. Develop strategies for the effective resource allocation in hydrogen consumption applications, considering performance and operational efficiency	2	1	-	-
<i>Hydrogen Re-fueling Systems</i>	6	11	-	-
PC26. Apply knowledge of hydrogen re-fueling systems to manage and maintain these systems safely and effectively, ensuring compliance with safety protocols and overseeing refueling operations	1	1	-	-
PC27. Analyze and assess the safety and efficiency of hydrogen re-fueling systems including storage at the refueling stations, making improvements as needed and ensuring compliance with safety standards.	1	2	-	-
PC28. Develop strategies for overseeing refueling processes, ensuring safety and efficiency in hydrogen re-fueling operations and compliance with relevant protocols.	1	2	-	-
PC29. Monitoring parameters like pressure, temperature during refueling operation	1	2	-	-
PC30. Monitoring Hydrogen detector / flame detectors and activate safety protocol in case of leakage/ emergency.	1	2	-	-
PC31. Ensure adherence to appropriate dress code by the operating personnel at the refueling station	1	2	-	-
NOS Total	35	40	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4006
NOS Name	Hydrogen Technology Overview and Operations Management
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	5
Credits	6
Version	1.0
Last Reviewed Date	30/11/2023
Next Review Date	29/11/2026
NSQC Clearance Date	30/11/2023

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HYC/N4007: Managing Hydrogen production system & Technologies

Description

This unit is about hydrogen technologies, hydrogen power energy systems, Hydrogen production and future trends and challenges. An individual after having undergone this module shall be able to decide on selection of appropriate technologies related to hydrogen production, scale of production, managing quality of hydrogen, role of feed water quality. Operation of Electrolyser and safety of the plant.

Scope

The scope covers the following :

- Perform activities related to Green Hydrogen technology selection, optimising production process, assessment of social and environmental impact and requirement of technology upgrading in the existing system
- Monitoring quality control of water and hydrogen produced
- Monitor Safety during production of hydrogen

Elements and Performance Criteria

Perform activities related to Green Hydrogen technology selection, optimising production process, assessment of social and environmental impact and requirement of technology upgrading in the existing system.

To be competent, the user/individual on the job must be able to:

- PC1.** Compare different types of electrolyzers their operation temperature, stack voltage efficiency etc.
- PC2.** Understand electrolytic hydrogen production, hydrogen re-electrification using fuel cell, hydrogen storage and converter technologies.
- PC3.** Familiarization with Hydrogen production methods like steam reforming, coal gasification and electrolysis of water and emerging methods like reforming ethanol and sugars, water biophotolysis
- PC4.** Understand thoroughly Water electrolysis principles and its electrical characteristics.
- PC5.** Fully understand electrolyzer design and construction including MOC.
- PC6.** Interpret Power and efficiency curves of various types of electrolyser, perform stack voltage efficiency calculations, determine/calculate system efficiency of electrolyser and calculate Hydrogen production rate.
- PC7.** determine the start, stop, adjust parameters, and monitor equipment performance to achieve optimal electrolysis
- PC8.** Start and monitor the power supply system, which include renewable energy sources

Monitoring quality control of water and hydrogen produced

To be competent, the user/individual on the job must be able to:

- PC9.** monitor and regulate the water supply to the electrolyzer
- PC10.** Monitor and regulate hydrogen production rate.

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- PC11.** control and monitor the water purification system, which involve filters, deionization units, or reverse osmosis systems, to maintain the required water quality including sources of impurities in water
- PC12.** operate gas purification systems to remove impurities from the hydrogen gas
- PC13.** determine gas separation processes, such as pressure swing adsorption (psa) or membrane separation, used to obtain pure hydrogen gas
- PC14.** monitor storage tanks or cylinders to ensure proper storage conditions for the produced hydrogen gas.
- PC15.** report logs, finding and fault to the supervisor/manager

Monitor Safety during production of hydrogen

To be competent, the user/individual on the job must be able to:

- PC16.** ensuring compliance with safety and quality standards during production of green hydrogen
- PC17.** report logs daily, report near miss, incidents, findings to the supervisor/manager.
- PC18.** ensuring compliance with safety and quality standards during production of green hydrogen
- PC19.** maintain & record of important parameters of electrolyzer like electrolyte flow, temp, current, h₂ flow rate, pressure parameters.
- PC20.** monitor proper periodic cip record also need to maintain & proper maintenance the rejected water from ro either through zero liquid discharge scheme.
- PC21.** apply generic standard operating procedures
- PC22.** apply basic knowledge of MSDS used for KOH, H₂, O₂
- PC23.** handle safe collection of laboratory samples
- PC24.** handle emergency actions, troubleshooting

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the basic principles and processes involved in hydrogen production
- KU2.** the properties and use of green hydrogen
- KU3.** Characteristics of hydrogen technologies
- KU4.** standard operating procedures for green hydrogen production
- KU5.** types of electrolyzers and their performance efficiency
- KU6.** importance of calibration of tools and instruments used in green hydrogen production
- KU7.** hydrogen output per unit of time
- KU8.** the block flow diagram and process flow diagram
- KU9.** basic heat & material balance
- KU10.** the basic PID with legends
- KU11.** organizational procedures for safe handling of hydrogen gas
- KU12.** reporting protocol and documentation required
- KU13.** relevant health and safety requirements applicable to the workplace
- KU14.** the importance of personal protective equipment used during the procedure performing
- KU15.** ways of efficiently managing wind and water in the process

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- KU16.** latest methods of energy and material conservation
- KU17.** categorization of waste into dry, wet, recyclable, non-recyclable and items of single-use plastics
- KU18.** the difference between recyclable/reusable and waste material
- KU19.** MSDS of all chemicals
- KU20.** PID interlocks emergency shut down actions
- KU21.** properly maintaining logbook & proper shift handover are practiced everyday very effectively
- KU22.** work permits are signed after ensuring safety & in time on daily basis
- KU23.** safe startup & shutdown of electrolyzers /compressors /h₂ deoxo etc. as per sop.
- KU24.** the concept of Mission Karmayogi
- KU25.** the concept of Swachhata Hi Seva

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate by writing
- GS2.** read instructions, guidelines/procedures and reports
- GS3.** communicate effectively and share the information efficiently
- GS4.** identify and report potential sources of danger
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of on time services
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform activities related to Green Hydrogen technology selection, optimising production process, assessment of social and environmental impact and requirement of technology upgrading in the existing system.</i>	13	7	-	-
PC1. Compare different types of electrolyzers their operation temperature, stack voltage efficiency etc.	2	-	-	-
PC2. Understand electrolytic hydrogen production, hydrogen re-electrification using fuel cell, hydrogen storage and converter technologies.	2	-	-	-
PC3. Familiarization with Hydrogen production methods like steam reforming, coal gasification and electrolysis of water and emerging methods like reforming ethanol and sugars, water biophotolysis	1	2	-	-
PC4. Understand thoroughly Water electrolysis principles and its electrical characteristics.	1	-	-	-
PC5. Fully understand electrolyzer design and construction including MOC.	2	-	-	-
PC6. Interpret Power and efficiency curves of various types of electrolyser, perform stack voltage efficiency calculations, determine/calculate system efficiency of electrolyser and calculate Hydrogen production rate.	2	-	-	-
PC7. determine the start, stop, adjust parameters, and monitor equipment performance to achieve optimal electrolysis	1	2	-	-
PC8. Start and monitor the power supply system, which include renewable energy sources	2	3	-	-
<i>Monitoring quality control of water and hydrogen produced</i>	10	17	-	-
PC9. monitor and regulate the water supply to the electrolyzer	1	2	-	-
PC10. Monitor and regulate hydrogen production rate.	1	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. control and monitor the water purification system, which involve filters, deionization units, or reverse osmosis systems, to maintain the required water quality including sources of impurities in water	2	3	-	-
PC12. operate gas purification systems to remove impurities from the hydrogen gas	2	3	-	-
PC13. determine gas separation processes, such as pressure swing adsorption (psa) or membrane separation, used to obtain pure hydrogen gas	2	3	-	-
PC14. monitor storage tanks or cylinders to ensure proper storage conditions for the produced hydrogen gas.	1	3	-	-
PC15. report logs, finding and fault to the supervisor/manager	1	2	-	-
<i>Monitor Safety during production of hydrogen</i>	17	18	-	-
PC16. ensuring compliance with safety and quality standards during production of green hydrogen	2	2	-	-
PC17. report logs daily, report near miss, incidents, findings to the supervisor/manager.	2	2	-	-
PC18. ensuring compliance with safety and quality standards during production of green hydrogen	2	2	-	-
PC19. maintain & record of important parameters of electrolyzer like electrolyte flow, temp, current, h ₂ flow rate, pressure parameters.	1	2	-	-
PC20. monitor proper periodic cip record also need to maintain & proper maintenance the rejected water from ro either through zero liquid discharge scheme.	2	2	-	-
PC21. apply generic standard operating procedures	2	2	-	-
PC22. apply basic knowledge of MSDS used for KOH, H ₂ , O ₂	2	2	-	-
PC23. handle safe collection of laboratory samples	2	2	-	-
PC24. handle emergency actions, troubleshooting	2	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	40	42	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HYC/N4007
NOS Name	Managing Hydrogen production system & Technologies
Sector	Hydrocarbon
Sub-Sector	Downstream
Occupation	Hydrogen Production, Storage & Transportation
NSQF Level	5
Credits	6.5
Version	1.0
Last Reviewed Date	30/11/2023
Next Review Date	29/11/2026
NSQC Clearance Date	30/11/2023

Qualification Pack

HYC/N9301: Working effectively in a team

Description

This unit covers basic practices that improve effectiveness of working in a team in an organizational set-up

Scope

The scope covers the following :

- Effective team work

Elements and Performance Criteria

Effective team work

To be competent, the user/individual on the job must be able to:

- PC1.** maintain clear communication with colleagues
- PC2.** pass on information to colleagues in line with organisational requirements
- PC3.** work in a team and support the team members
- PC4.** work in ways that show respect to colleagues
- PC5.** fulfil commitments made to colleagues
- PC6.** inform team members timely, if timelines can't be met
- PC7.** take the necessary initiatives to resolve the issues while working in team

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization policies and procedures related to team performance
- KU2.** the importance of effective communication and establishing good working relationships with colleagues
- KU3.** Deeper understanding of actions & consequences of gendered behaviour
- KU4.** the importance of creating an environment of trust and mutual respect
- KU5.** the implications of own work on the work and schedule of others
- KU6.** the standard practice in organization w.r.t communication at various levels
- KU7.** the instructions at individual level or at group level
- KU8.** individual work responsibility and corrective measures required to complete task in time
- KU9.** the importance of personal qualities like discipline and confidence to achieve success in work assigned

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** communicate effectively in writing
- GS2.** read instructions, guidelines/procedures
- GS3.** communicate information effectively
- GS4.** make decisions on a suitable course of action or response keeping in view resource utilization while meeting commitments
- GS5.** how to plan the work to meet the deadline
- GS6.** the importance of consistent and reliable services for customer satisfaction
- GS7.** apply problem solving approaches in different situations
- GS8.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Effective team work</i>	20	30	-	-
PC1. maintain clear communication with colleagues	2	4	-	-
PC2. pass on information to colleagues in line with organisational requirements	3	5	-	-
PC3. work in a team and support the team members	3	4	-	-
PC4. work in ways that show respect to colleagues	3	4	-	-
PC5. fulfil commitments made to colleagues	3	5	-	-
PC6. inform team members timely, if timelines can't be met	3	4	-	-
PC7. take the necessary initiatives to resolve the issues while working in team	3	4	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HYC/N9301
NOS Name	Working effectively in a team
Sector	Hydrocarbon
Sub-Sector	Generic
Occupation	Generic, Generic
NSQF Level	4
Credits	2.5
Version	5.0
Last Reviewed Date	27/08/2024
Next Review Date	26/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment

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- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms

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- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services

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- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of aggregate marks to successfully clear the assessment.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HYC/N4001.Hydrogen Safety, Security and Health Management Procedures	35	60	-	-	95	30
HYC/N4006.Hydrogen Technology Overview and Operations Management	35	40	-	-	75	30
HYC/N4007.Managing Hydrogen production system & Technologies	40	42	-	-	82	20
HYC/N9301.Working effectively in a team	20	30	-	-	50	10
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	10
Total	150	202	-	-	352	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure
NOS	National Occupational Standard(s)
OS	Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
KU	Knowledge and Understanding
GS	Generic Skills
SOP	Standard operating procedure

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.